Some Thoughts on DICE
(mostly from Spring 2019)

Converting DICE to Use a 1-Year Time Step (or choice of time step)

EXOGENOUS DATA

Population

Key lines of code in DICE2016R2

pop0	Initial world population 2015 (millions)	/7403/
popadj	Growth rate to calibrate to 2050 pop projection	/0.134/
popasym	Asymptotic population (millions)	/11500/
l(t)	Level of population and labor	

```
l("1") = pop0;
loop(t, l(t+1)=l(t)*(popasym/L(t))**popadj ;);
```
(note: t is an index indicating the time period)

Interpretation

Replacing 'l' and 'L' with 'pop' and putting the indicator for the time period in a subscript, we can rewrite these equations as

$$pop_1 = pop0$$

$$pop_{t+1} = pop_t * \left(\frac{popasym}{pop_t}\right)^{popadj}$$

This describes a sigmoid function, which can be solved to yield:

$$pop_t = \left(pop_1^{popadj} * e^{-popadj*(t-1)} + popasym^{popadj} * \left(1 - e^{-popadj*(t-1)}\right)\right)^{\frac{1}{popadj}}$$

where t is the time period, with t = 1 in 2015, 2 in 2020, etc.

Nordhaus provides the value of popasym as 11,500 and popadj as 0134. The value for popadj is for 5-year time periods. To convert this into a value for a 1-year time step, it is possible to show the following:

$$popadj_1 = \ 1 - \left(1 - popadj_{tstep}\right)^{\frac{1}{tstep}}$$

where tstep, the length of the period in years, is a positive integer

This can be solved with t = 5 to determine:

$$popadj_1 \approx 0.028364,$$

Furthermore, we can rewrite this equation as:

$$popadj_{tstep} = 1 - (1 - popadj_1)^{tstep}$$

This can be used to generate values of $popadj_{tstep}$ for any choice of tstep, where tstep is a positive integer.

TFP

Key lines of code in DICE2016R2

a0	Initial level of total factor productivity	/5.115/[1]
ga0	Initial growth rate for TFP per 5 years	/0.076/
dela	Decline rate of TFP per 5 years	/0.005/
ga(t)	Growth rate of productivity from	
al(t)	Level of total factor productivity	

```
ga(t) = ga0*exp(-dela*5*((t.val-1)));
al("1") = a0; loop(t, al(t+1)=al(t)/((1-ga(t))););
```

(note: t is an index indicating the time period; t.val is the value for time period, with t.val = 1 in 2015, 2 in 2020, etc.)

Interpretation

Combining the two equations above, noting that t = t.val, and putting the indicator for the time period in a subscript, we get

$$al_1 = a0$$

$$al_{t+1} = \frac{al_t}{1 - ga0 * e^{(-dela*5*(t-1))}}$$

where t is the time period, with t = 1 in 2015, 2 in 2020, etc.

---

[1] We found a slight discrepancy in this value by noting that the equation for gross output, YGROSS(t) is given as:

```
YGROSS(t) = (al(t)*(L(t)/1000)**(1-GAMA))*(K(t)**GAMA),
```

which can be rearranged to yield:

```
al(t) = YGROSS(t)/((L(t)/1000)**(1-GAMA))*(K(t)**GAMA))
```

For t = 1, DICE provides the following values: YGROSS = 105.5, L = 7403, K = 223, and gama = 0.3. Solving for al yields a value of 5.131, which is slightly lower than that for a0, which is given as 5.115. We have adjusted the code to directly calculate a0 so as to ensure consistency between these initial values.

Subtracting al_t from each side yields:

$$al_{t+1} - al_t = \frac{al_t}{1 - ga0 * e^{(-dela*5*(t-1))}} - al_t$$

$$al_{t+1} - al_t = al_t * \frac{ga0 * e^{(-dela*5*(t-1))}}{1 - ga0 * e^{(-dela*5*(t-1))}}$$

This is another form of a sigmoid function, which can be solved to yield:

$$al_t = a0 * \left(\frac{1 - ga0 * e^{(-dela*5*(t-1))}}{1 - ga0}\right)^{\frac{1}{dela*5}}$$

When t =1, this reduces to $al_t = a0$, as expected. As t approaches ∞, we have:

$$al_t \to a0 * \left(\frac{1}{1 - ga0}\right)^{\frac{1}{dela*5}}$$

Thus, the asymptotic value of TFP is a function of the initial value, a0, the initial growth rate, ga0, and the rate of decline of the growth rate, dela.

Here we need to look at the values of dela and ga0. Though dela is specified as being the decline rate of TFP for five years, in looking at the equation for al_{t+1}, it is clear that it is an annual value, which is multiplied by 5, the length of the time step. Thus, there is no need to modify its value, but it is necessary to replace the number 5 in the code with tstep. On the other hand, ga0 does need to be adjusted to reflect the shift to a 1-year time step. Making the reasonable assumption that the asymptotic value of al_t as t → ∞ should be independent of the time step, we can show that:

$$ga0_1 = 1 - \left(1 - ga0_{tstep}\right)^{\frac{1}{tstep}}$$

where tstep, the length of the period in years, is a positive integer

This can be solved with t = 5 and $ga0_5$ = 0.076 to determine:

$$ga0_1 \approx 0.015684$$

Furthermore, we can rewrite this equation as:

$$ga0_{tstep} = 1 - (1 - ga0_1)^{tstep}$$

This can be used to generate values of $ga0_{tstep}$ for any choice of tstep, where tstep is a positive integer.

Land Use Emissions

Key lines of code in DICE2016R2

eland0	Carbon emissions from land 2015 (GtCO2 per year)	/2.6/
deland	Decline rate of land emissions (per period)	/0.115/
etree(t)	Emissions from deforestation	

```
etree(t) = eland0*(1-deland)**(t.val-1);
```

(note: t is an index indicating the time period; t.val is the value for time period, with t.val = 1 in 2015, 2 in 2020, etc.)

Interpretation

Noting that t = t.val, and putting the indicator for the time period in a subscript, we get

$$etree_1 = eland0$$

$$etree_{t+1} = etree_t * \ (1 - deland)$$

Note that the annual values, etree(t), are multiplied by 5 later in the code to get the total amount added to the atmosphere over each 5-year period.

This represents a simple geometric decline in emissions over time, where deland represents the decline over one period using a 5-year time step. As with popasym, to convert this into a value for a 1-year time step, it is possible to show the following:

$$deland_1 = \ 1 - \left(1 - deland_{tstep}\right)^{\frac{1}{tstep}}$$

where tstep, the length of the period in years is a positive integer

This can be solved with t = 5 to determine:

$$deland_1 \approx 0.024137$$

Again, we can rewrite this equation as:

$$deland_{tstep} = \ 1 - (1 - deland_1)^{tstep}$$

to generate values of $deland_{tstep}$ for any choice of tstep, where tstep is a positive integer.

Carbon Intensity

Key lines of code in DICE2016R2

gsigma1	Initial growth of sigma (per year)	/-0.0152/
dsig	Decline rate of decarbonization (per period)	/-0.001/
e0	Industrial emissions 2015 (GtCO2 per year)	/35.85/
miu0	Initial emissions control rate for base case 2015	/0.03/

q0 Initial world gross output 2015 (trill 2010 USD) /105.5/
sig0 Carbon intensity 2010 (kgCO2 per output 2005 USD 2010)
sig0 = e0/(q0*(1-miu0))
gsig(t) Change in sigma (cumulative improvement of energy efficiency)
sigma(t) CO2-equivalent-emissions output ratio

```
sig("1") = gsigma1;    loop(t, gsig(t+1) = gsig(t)*((1+dsig)**tstep) ;);
sigma("1") = sig0;     loop(t, sigma(t+1) = (sigma(t)*exp(gsig(t)*tstep)););
```

(note: t is an index indicating the time period; t.val is the value for time period, with t.val = 1 in 2015, 2 in 2020, etc.; tstep is the length of the time step, which is given the value 5)

Interpretation

Noting that t = t.val, and putting the indicator for the time period in a subscript, we get:

$$gsig_1 = gsigma1$$

$$gsig_{t+1} = gsig_t * (1 + dsig)^{tstep}$$

$$sigma_1 = sig0$$

$$sigma_{t+1} = sigma_t * e^{gsig_t * tstep}$$

Here we can notice that since dsig < 0, the pattern for computing $gsig_{t+1}$ is the same as we saw for land use emissions. Also, even though the description of dsig says it is per period, the equation for gsig actually implies that it is per year. Thus, we do not need to convert it from a 5-year time step to a 1-year time step. We can still use the method described for popyasym and deland to calculate values for dsig for time steps greater than one year, in which case we can also drop the inclusion of tstep in the equation for $gsig_{t+1}$.

Other Forcing

Key lines of code in DICE2016R2

fex0 2015 forcings of non-CO2 GHG (Wm-2) /0.5/
fex1 2100 forcings of non-CO2 GHG (Wm-2) /1.0/

```
forcoth(t) = fex0+ (1/17)*(fex1-fex0)*(t.val-1)$(t.val lt 18)+ (fex1-fex0)$(t.val ge 18);
```

(note: t is an index indicating the time period; t.val is the value for time period, with t.val = 1 in 2015, 2 in 2020, etc.)

Interpretation

This equation implies that forcoth(t) increases linearly from the value given by fex0 to that given by fex1 over the first 17 periods, or 17*5 = 85 years, and then remains constant at the value given by fex1.

All we need do here is to rewrite the equation to incorporate the time step. Noting that t = t.val, and putting the indicator for the time period in a subscript, we get:

$$forcoth_t = fex0 + \left(\frac{tstep}{85}\right) * (fex1 - fex0) * (t-1) \; if \left(t \le \left(\frac{85}{tstep}\right)\right)$$
$$+ (fex1 - fex0) \; if \left(t > \left(\frac{85}{tstep}\right)\right)$$

where tstep, the length of the period in years, is a positive integer less than or equal to 85

Backstop Technology Price

Key lines of code in DICE2016R2

pback	Cost of backstop 2010$ per tCO2 2015	/550/
gback	Initial cost decline backstop cost per period	/0.025/
pbacktime(t)	Backstop price	

pbacktime(t)=pback*(1-gback)**(t.val-1);

(note: t is an index indicating the time period; t.val is the value for time period, with t.val = 1 in 2015, 2 in 2020, etc.)

Interpretation

This is the same approach as used for land use emissions. Following the same logic, we have:

$$pbacktime_1 = pback$$

$$pbacktime_{t+1} = pbacktime_t * \left(1 - gback_{tstep}\right)$$

$$gback_{tstep} = 1 - (1 - gback_1)^{tstep}$$

$$gback_1 \approx 0.005051$$

where tstep, the length of the period in years, is a positive integer

MODULES

Carbon Cycle Model

Key lines of code in DICE2016R2

** Carbon cycle

* Initial Conditions

mat0	Initial Concentration in atmosphere 2015 (GtC)	/851/
mu0	Initial Concentration in upper strata 2015 (GtC)	/460/
ml0	Initial Concentration in lower strata 2015 (GtC)	/1740/
mateq	Equilibrium concentration atmosphere (GtC)	/588/
mueq	Equilibrium concentration in upper strata (GtC)	/360/
mleq	Equilibrium concentration in lower strata (GtC)	/1720/

* Flow paramaters

```
b12     Carbon cycle transition matrix                         /0.12/
b23     Carbon cycle transition matrix                         /0.007/
* These are for declaration and are defined later
b11     Carbon cycle transition matrix
b21     Carbon cycle transition matrix
b22     Carbon cycle transition matrix
b32     Carbon cycle transition matrix
b33     Carbon cycle transition matrix
* Parameters for long-run consistency of carbon cycle
b11 = 1 - b12;
b21 = b12*MATEQ/MUEQ;
b22 = 1 - b21 - b23;
b32 = b23*mueq/mleq;
b33 = 1 - b32;

mmat(t+1)..        MAT(t+1) =E= MAT(t)*b11 + MU(t)*b21 + (E(t)*(5/3.666));
mml(t+1)..         ML(t+1) =E= ML(t)*b33  + MU(t)*b23;
mmu(t+1)..         MU(t+1) =E= MAT(t)*b12 + MU(t)*b22 + ML(t)*B32;
```

(note: t is an index indicating the time period; E is the sum of industrial and land use emissions in $GtCO_2$/year, and 3.666 is a factor for converting GtC to $GtCO_2$)

Interpretation

As noted in Nordhaus and Storcz (2013, p.16), the carbon cycle model is a three-reservoir model, with inputs from current emissions and exchanges between the three reservoirs. This can be seen more clearly if we rewrite the equations above as:

$$MAT_{t+1} = MAT_t + E_t * \frac{5}{3.666} - b12 * \left(MAT_t - MU_t * \frac{mateq}{mueq}\right)$$

$$MU_{t+1} = MU_t + b12 * \left(MAT_t - MU_t * \frac{mateq}{mueq}\right) - b23 * \left(MU_t - ML_t * \frac{mueq}{mleq}\right)$$

$$ML_{t+1} = ML_t + b23 * \left(MU_t - ML_t * \frac{mueq}{mleq}\right)$$

The second term in the equation for MAT shows that the carbon in the atmosphere increases with CO_2 emissions, E_t, which are multiplied by 5 to reflect the time step (recall that 3.666 is simply a factor for converting GtC to $GtCO_2$). The third term describes the transfer of carbon between the atmosphere and the upper ocean based on equilibrium dynamics. Given a positive value for b12, if the amount of carbon in the atmosphere exceeds that in the upper ocean, adjusted by their relative equilibrium levels, there will be a net transfer of carbon from the atmosphere to the upper ocean, and vice versa. This transfer is made clear by the appearance of the same term in the equation for MU, but with an opposite sign. A similar term for the exchange of carbon between the upper and lower ocean appears in the equations for MU and ML, again with opposite signs in the two equations.

In order to convert this set of equations from a 5-year to a 1-year (or any) time step, we need to adjust the emissions and the values of the scalars b12 and b23, which control the rates of transfer between the reservoirs. The first is easily done by replacing the value 5 with a parameter, tstep, in the equation for MAT, where tstep is a positive integer. For b12 and b23, however, unlike the time-series parameters, there is not a simple way to convert the underlying scalars. This is because of the nature of the set of equations and because these scalars have been calibrated to existing carbon-cycle models and historical data assuming a 5-year time step.

To solve this problem, we have recalculated the values for b12 and b23 to a 1-year time step as follows.

1. We ran the DICE model, using a 5-year time step and default scalars, but with no climate damages and no emission controls. From this we took the values for total CO_2 emissions, other forcing, MAT, MU, and ML for each 5-year time step.
2. For the "in-between" years, i.e. the years between the 5-year time steps, we used the value at the start of the time period, e.g. the values for 2016, 2017, 2018, and 2019 are the same as that provided by DICE for 2015. This is consistent with how DICE works with a multi-year time step.
3. We then used the equations above to calculate the values of MAT, MU, and ML.
4. We adjusted the values of b12 and b23 to minimize the sum of the squared differences between the values calculated for MAT, MU, and ML and those calculated in step 1 above for each 5-year time step.

The resulting values for b12 and b23 are 0.023928 and 0.001389, respectively.

In the model itself, depending on the value of tstep, the model will select the correct values for b12 and b23. For the moment, this only works with tstep = 1 or 5; for other values of tstep, we would need to do additional recalibration.

Climate Model

Key lines of code in DICE2016R2

t2xco2	Equilibrium temp impact (°C per doubling CO2)	/3.1/
fco22x	Forcings of equilibrium CO2 doubling (Wm-2)	/3.6813/
tocean0	Initial lower stratum temp change (°C from 1900)	/0.0068/
tatm0	Initial atmospheric temp change (°C from 1900)	/0.85/
c1	Climate equation coefficient for upper level	/0.1005/
c3	Transfer coefficient upper to lower stratum	/0.088/
c4	Transfer coefficient for lower level	/0.025/

```
force(t)..          FORC(t) =E= fco22x * ((log((MAT(t)/588.000))/log(2))) +
                    forcoth(t);
tatmeq(t+1)..       TATM(t+1) =E= TATM(t) + c1 * ((FORC(t+1) -
                    (fco22x/t2xco2)*TATM(t))-(c3*(TATM(t)-TOCEAN(t))));
toceaneq(t+1)..     TOCEAN(t+1) =E= TOCEAN(t) + c4*(TATM(t)-TOCEAN(t));
```

(note: t is an index indicating the time period; forcoth is the forcing from non-CO_2 gases in watts/m^2)

Interpretation
The climate model is a two-box model, where TATM(t) and TOCEAN(t) represent respectively the mean surface temperature (atmosphere and upper ocean) and the temperature of the deep oceans. The above equations can be rewritten for somewhat greater clarity as:

$$FORC_t = fco22x * \left(\frac{log\left({}^{MAT_t}/_{mateq}\right)}{\log(2)} \right) + forcoth_t$$

$$TATM_{t+1} = TATM_t + c1 * \left(FORC_{t+1} - \frac{fco22x}{t2xco2} * TATM_t \right) - c1 * c3 * (TATM_t - TOCEAN_t)$$

$$TOCEAN_{t+1} = TOCEAN_t + c4 * (TATM_t - TOCEAN_t)$$

The equation for FORC includes two components: 1) forcing by CO_2 in the atmosphere and 2) forcing by non-CO_2 gases in the atmosphere. The latter is provided exogenously. The former shows that the CO_2 forcing increases as the logarithm of the amount of CO_2 in the atmosphere, with fco22x indicating the level of forcing associated with a doubling of the amount of CO_2 in the atmosphere from the pre-industrial "equilibrium" level.

The atmospheric temperature, TATM, changes as a result of two processes. The first relates the level of current forcing to the level of forcing that would be associated with the current value of TATM, if we were in long-term equilibrium. Since c1 is positive, if the current forcing is larger than this long-term equilibrium value, TATM will increase, and vice-versa. The second process relates to the difference between TATM and the temperature of the deep ocean, TOCEAN; this same process is also seen in the equation for TOCEAN. since c1, c3, and c4 are all positive, if TATM is greater than TOCEAN, this will result in a decrease in the former and an increase in the latter, and vice versa. The differences in the values of c1*c3 (0.1005*0.088 = 0.008844) differs from that of c4 (0.025) reflect the different rates at which heat translates into temperature changes. *The fact that the value of c1*c3 is less than c4 seems counter-intuitive in that this would seem to imply that the heat capacity of the atmosphere and the upper ocean is greater than that of the deep ocean. This was not always the case with DICE. I wrote Nordhaus about this. He responded:*

> *"The other question relates to the coefficients v the size of the reservoirs. In the early years, the calibration had been designed to match reservoir size. However, we found that this led to poor calibration with major model results. In the 2016 model, we began with the Archer et al.[2] carbon cycle calibration. Then, based on*

[2] Archer, David, Michael Eby, Victor Brovkin, Andy Ridgwell, Long Cao, Uwe Mikolajewicz, Ken Caldeira, et al. 2009. "Atmospheric Lifetime of Fossil Fuel Carbon Dioxide." *Annual Review of Earth and Planetary Sciences* 37 (1): 117–34. https://doi.org/10.1146/annurev.earth.031208.100206.

> *that, we calibrated the climate model to transient and equilibrium climate sensitivity (roughly 1.7 and 3.1) (personal correspondence 15 Feb 2019).”*

As with the carbon cycle model, there is not a simple way to convert the underlying scalars because of the nature of the set of equations and because these scalars have been calibrated to existing climate models and historical data assuming a 5-year time step. Following a similar procedure as given above, though, we were able to come up with estimates of c1, c3, and c4 for a 1-year time step, which are 0.022324, 0.090479, and 0.004992. *We are still left with the problem that c1*c3 = 0.002324, is less than c4.*

In the model itself, depending on the value of tstep, the model will select the correct values for c1, c3, and c4. For the moment, this only works with tstep = 1 or 5; for other values of tstep, we would need to do additional recalibration.

Abatement Costs in DICE

There are a number of issues related to the abatement costs in DICE.

DICE 2016-R2

expcost2	Exponent of control cost function	/2.6/
pbacktime(t)	Backstop price	
pback	Cost of backstop 2010$ per tCO2 2015	/550/
gback	Initial cost decline backstop cost per period	/0.025/
sigma(t)	CO2-equivalent-emissions output ratio (GTCO2/trillion USD)	
gsigma1	Initial growth of sigma (per year)	/-0.0152/
dsig	Decline rate of decarbonization (per period)	/-0.001/
e0	Industrial emissions 2015 (GtCO2 per year)	/35.85/
miu0	Initial emissions control rate for base case 2015	/0.03/
q0	Initial world gross output 2015 (trill 2010 USD)	/105.5/
cost1(t)	Adjusted cost for backstop	

```
pbacktime(t)=pback*(1-gback)**(t.val-1);
sig0 = e0/(q0*(1-miu0));
gsig("1")=gsigma1; loop(t,gsig(t+1)=gsig(t)*((1+dsig)**tstep) ;);
sigma("1")=sig0; loop(t,sigma(t+1)=(sigma(t)*exp(gsig(t)*tstep)););
cost1(t) = pbacktime(t)*sigma(t)/expcost2/1000;
```

YGROSS(t) Gross world product GROSS of abatement and damages (trillions 2005 USD per year)[3]

ABATECOST(t) Total cost of emissions reductions (trillions 2005 USD per year)[4]

MCABATE(t) Marginal cost of emissions reductions (2005$ per ton CO2)[5]

ABATEEQ(t) Cost of emissions reductions equation

MCABATEEQ(t) Equation for MC abatement

```
abateeq(t)..        ABATECOST(t) =E= YGROSS(t) * cost1(t) *
(MIU(t)**expcost2);
mcabateeq(t)..      MCABATE(t) =E= pbacktime(t) * MIU(t)**(expcost2-1);
```

(note: t is an index indicating the time period; t.val is the value for time period, with t.val = 1 in 2015, 2 in 2020, etc.; MIU(t) is a choice variable so does not have an equation)

Interpretation

Nordhaus (2018, p. 337) states:

> "The abatement cost function, $\Lambda(t)$ in equation (2) above, is the fraction of output devoted to reducing CO_2 emissions (perhaps 3 percent). It was recalibrated to the abatement cost functions of other IAMs as represented in the modeling uncertainty project (MUP) study (Gillingham et al. forthcoming). The resulting abatement function has slightly higher costs than earlier estimates. The abatement-cost function takes the form $\Lambda(t) = \theta_1(t)\mu(t)^{\theta_2}$, where $\theta_1(t) = 0.0741 \times$

---

[3] Although the code says 2005 USD, we assume he means 2010 USD given the definition of other monetary values.

[4] See previous footnote.

[5] See previous footnote.

$0.0904^{(t-1)}$ and $\theta_2 = 2.6$.[6] The interpretation here is that at zero emissions for the first period, abatement is 7.41 percent of output. That percentage declines at 2 percent per year. The abatement-cost function is highly convex, reflecting the sharp diminishing returns to reducing emissions.

The model assumes the existence of a "backstop technology," which is a technology that produces energy services with zero greenhouse gas (GHG) emissions ($\mu = 1$). The backstop price in 2020 is $550 per ton of CO_2-equivalent, and the backstop cost declines at 1/2 percent per year[7]. Additionally, it is assumed that there are no "negative emissions" technologies initially, but that negative emissions are available after 2150. The existence of negative-emissions technologies ($\mu > 1$) is critical to reaching low-temperature targets, as described below."

There are a few things to note here:

- costl(t) is not really the adjusted cost for the backstop, but rather the fraction of gross output spent on abatement when MIU(t) = 1, i.e., there are zero emissions.
- The total cost of emissions reduction equation can be rewritten as ABATECOST(t) =E= YGROSS(t)*sigma(t)/1000 * (pbacktime(t)/expcost2)*MIU(t)**expcost2, which is just the total amount of uncontrolled emissions times the definite integral of the marginal cost of emissions reduction equation evaluated from 0 to MIU(t), which it should be.
- The reference to Gillingham et al, forthcoming, is to a paper published in the *Journal of the Association of Environmental and Resource Economists*. This paper, published as Gillingham et al (2018), presents no analysis related to marginal abatement costs. An earlier version, Gillingham et al (2015), presents a single figure (Figure 4, p.26) illustrating the percentage reduction in emissions as a function of carbon price for 6 different IAMs, including an earlier version of DICE. It does not provide sufficient detail, however to determine how Nordhaus did his calibration. Also, note that only one of the models showed percentage reductions greater than 100%.
- If we focus on the marginal cost of emissions reduction equation, we see that this is a function of the fraction of emissions abated, not the absolute amount. We discuss the implications of this in the next section.
- He does not provide separate cost functions for negative emissions technologies. We discuss the implications of this in the following section.

---

[6] The expression $\theta_1(t) = 0.0741 \times 0.0904^{(t-1)}$ does not appear in the model code. Rather, based on the cost of emissions reduction equation $\theta_1(t)$ is given by costl(t). Letting t = 1 and MIU(1) = 1, we have costl(1) = pbacktime(1)*sigma(1)/expcost2/1000 = pback*sig0/expcost2/1000 = pback*e0/(q0*(1-miu0))/expcost2/1000 = 550*35.85/(105.5*(1-0.03))/2.6/1000 = 0.0741. This matches his statement that "at zero emissions for the first period, abatement is 7.41 percent of output." The value 0.0904 is more difficult to reconstruct as it is a function of the rates at which the price of the backstop technology and the carbon intensity of the economy fall. It looks like he may have misplaced the decimal point here, though. A 2 percent per year decline over 5 years, the time step of his model, would yield a value of $(1-0.02)^5 = 0.904$, not 0.0904.

[7] 550 matches the default value of pback, which, in the model, actually reflects the price in 2015, not 2020. He multiples the value of ½ percent per year to get the default value of gback, 0.025.

The same basic formulation for the abatement cost function has been used for all versions of DICE. Nordhaus and Stzorc (2013), in describing DICE 2013R, present the same formulation, with the same parameterization for θ_2. On p.13, the provide a bit more detail on the Backstop technology:

> "The DICE-2013R model explicitly includes a backstop technology, which is a technology that can replace all fossil fuels. The backstop technology could be one that removes carbon from the atmosphere or an all-purpose environmentally benign zero-carbon energy technology. It might be solar power, or carbon-eating trees or windmills, or some as-yet undiscovered source. The backstop price is assumed to be initially high and to decline over time with carbon-saving technological change.
>
> In the full regional model, the backstop technology replaces 100 percent of carbon emissions at a cost of between \$230 and \$540 per ton of CO 2 depending upon the region in 2005 prices. For the global DICE-2013R model, the 2010 cost of the backstop technology is \$344 per ton CO 2 at 100% removal. The cost of the backstop technology is assumed to decline at 0.5% per year. The backstop technology is introduced into the model by setting the time path of the parameters in the abatement-cost equation (6) so that the marginal cost of abatement at a control rate of 100 percent is equal to the backstop price for a given year."

DICE-92-94: cost embedded in net output equation

B1	Intercept control cost function	/.0686/
B2	Exponent of control cost function	/2.887/
Cost:	B1*(MIU(T)**B2)	

Nordhaus (1992)

- p.13: total cost equation (II-10), $TC(t) = Q(t)b_1\mu(t)^{b2}$
- p.22: $b_1 = 0.0686$, $b_2 = 2.887$
- pp.46 - 51: Cost of reduction:

Nordhaus (1994)

- pp.59-65: pretty much same discussion as in Nordhaus (1992).
- Both point to Nordhaus (1991) for data for parameterization

DICE-99: introduce changing cost over time, which actually increases; change exponent

COST10	Intercept control cost function	/.03/
COST2	Exponent of control cost function	/2.15/
dmiufunc	Decline in cost of abatement function (per decade)	/-.08/
decmiu	Change in decline of cost function	/.005/
cost1(t)	cost function for abatement – fraction of global gross output	

```
gcost1(T)=dmiufunc*EXP(-decmiu*10*(ORD(T)-1));
cost1("1")=cost10; LOOP(T, cost1(T+1)=cost1(T)/((1+gcost1(T+1)));););
cost1(t)*(MIU(T)**cost2))
```

cost(t) = y.l(t)*(cost1(t)*(miu.l(t)**cost2));

"The coefficients b 1 (t) and b 2 and those of the quadratic damage function were set so that the optimal carbon tax and emissions control rates in DICE-99 matched the projections of these variables in the optimal run of RICE-99 (Nordhaus and Boyer 2000, p.104)". In RICE-99, there are not explicit regional cost equations. Rather, these fall out of the structure of the model in estimating the demand for carbon-energy and the calibration of the model to historic data, including the effect of a carbon tax of $50. Note that some of this gets conflated with the inclusion of a Hotelling effect due to the limited ultimate supply of carbon-energy.

DICE-08: introduce explicit backstop technology, the cost of which declines over time; changed exponent

EXPCOST2	Exponent of control cost function	/2.8/
PBACK	Cost of backstop 2005 000$ per tC 2005	/1.17/ = 1170
BACKRAT	Ratio initial to final backstop cost	/2/
GBACK	Initial cost decline backstop pc per decade	/.05/
COST1(t)	Adjusted cost for backstop	
GCOST1	Growth of cost factor	

```
cost1(T) = (PBACK*SIGMA(T)/EXPCOST2)* ((BACKRAT-1+ EXP (-gback*
        (ORD(T)-1)))/BACKRAT);
ABATECOST(T) =E= (PARTFRACT(T)**(1-expcost2))
        *YGROSS(T)*(cost1(t)*(MIU(T)**EXPcost2));
```

Nordhaus (2008, p.42)

"The abatement-cost equation is a reduced-form-type model in which the costs of emissions reductions are a function of the emissions-reduction rate, $\mu(t)$. The abatement cost function assumes that abatement costs are proportional to global output and to a polynomial function of the reduction rate. The cost function is estimated to be highly convex, indicating that the marginal cost of reductions rises from zero more than linearly with the reductions rate.

A new feature of the DICE-2007 model is that it explicitly includes a backstop technology, which is a technology that can replace all fossil fuels. The backstop technology could be one that removes carbon from the atmosphere or an all-purpose environmentally benign zero-carbon energy technology. It might be solar power, or nuclear-based hydrogen, or some as-yet-undiscovered source. The backstop price is assumed to be initially high and to decline over time with carbon-saving technological change. The backstop technology is introduced into the model by setting the time path of the parameters in the abatement-cost equation (A.6) so that the marginal cost of abatement at a control rate of 100 percent is equal to the backstop price for each year.[4] (p.42)" fn4: The abatement-cost function is calibrated to a survey of estimates of abatement-cost functions, as well as estimates made by the MiniCam (Edmonds 2007). See the discussion later in this chapter for a further description.

"The basic functional form for the abatement-cost function follows the structure assumed in the earlier DICE models. However, the structure has been reformulated over time to correct for an earlier modeling mistake. The implicit specification in the DICE model is that there is a "backstop technology." As noted earlier, this is a technology that can replace all carbon-emitting processes at a relatively high cost; that is, the backstop technology takes over when the emissions-control rate is 100 percent. The prior version used a functional form that implicitly and mistakenly assumed that the cost of the backstop technology increased over time.

The new version redefines the emissions-reduction equations by calibrating them to an explicit price and time profile of the backstop technology. The calibration of the new emissions-cost function is based on recent modeling efforts that calculate the cost of deep emissions cuts, the IPCC special report on sequestration (IPCC 2005), the IPCC Fourth Assessment Report, as well as modeling estimates provided by Jae Edmonds. In the new model, the cost of the backstop technology starts around $1,200 per metric ton of carbon and declines to $950 per metric ton by 2100. (p.52)"

Cost of backstop falls to ½ of initial value over time

DICE2013R: allow control rate to go to 1.2 after 2150; backstop price now expressed in terms of tCO₂, not CO; slight increase in initial cost from DICE-08

```
expcost2      Exponent of control cost function            /2.8/
pback         Cost of backstop 2005$ per tCO2 2010         /344/ = 344*44/12=1261 per tCO
gback         Initial cost decline backstop cost per period  /.025/
gcost1        Growth of cost factor
cost1(t)      cost1(t)
pbacktime(t)  Backstop price
pbacktime(t)=pback*(1-gback)**(t.val-1);
cost1(t) = pbacktime(t)*sigma(t)/expcost2/1000;
ABATECOST(t) =E= YGROSS(t) * cost1(t) * (MIU(t)**expcost2) * (partfract(t)**(1-
        expcost2));
MCABATE(t) =E= pbacktime(t) * MIU(t)**(expcost2-1);
```

Nordhaus and Stzorc (2013), present the same formulation, with the same parameterization for θ_2 as in DICE-08. The initial cost of the backstop technology changed, even after converting from tC to tCO_2, probably reflecting the change in the base year. On p.13, the provide a bit more detail on the backstop technology:

"The DICE-2013R model explicitly includes a backstop technology, which is a technology that can replace all fossil fuels. The backstop technology could be one that removes carbon from the atmosphere or an all-purpose environmentally benign zero-carbon energy technology. It might be solar power, or carbon-eating trees or windmills, or some as-yet undiscovered source. The backstop price is assumed to be initially high and to decline over time with carbon-saving technological change.

In the full regional model, the backstop technology replaces 100 percent of carbon emissions at a cost of between \$230 and \$540 per ton of CO_2 depending upon the region in 2005 prices. For the global DICE-2013R model, the 2010 cost of the backstop technology is \$344 per ton CO_2 at 100% removal. The cost of the backstop technology is assumed to decline at 0.5% per year. The backstop technology is introduced into the model by setting the time path of the parameters in the abatement-cost equation (6) so that the marginal cost of abatement at a control rate of 100 percent is equal to the backstop price for a given year."

No limit on how far the cost of the backstop technology can fall

DICE 2016-R2: change exponent to 2.6; change initial cost of backstop to 550
No changes in formulation, but changes in both exponent and initial cost of backstop, the latter perhaps due to changing base year to 2015 and currency year to 2010

Using a Marginal Cost curve that is a function of abatement fraction

As noted above the marginal cost of emissions reduction equation in DICE is a function of the fraction of emissions abated. We can write this as MCABATE(MIU(t)) = pbacktime(t) * MIU(t)**(expcost2-1), where: 1) the default value of pbacktime(t) starts at a value of \$550 per ton of CO_2-equivalent in 2015, and declines at ½ percent per year, 2) MIU(t) is a choice variable in the optimization, and 3) expcost is a constant with a default value of 2.6.

The figure below plots MCABATE(t) as a function of MIU(t) for every 50 years over the period 2015 to 2315. This shows the steadily declining costs over time for each level of control, which is solely a function of the declining cost of the backup technology.

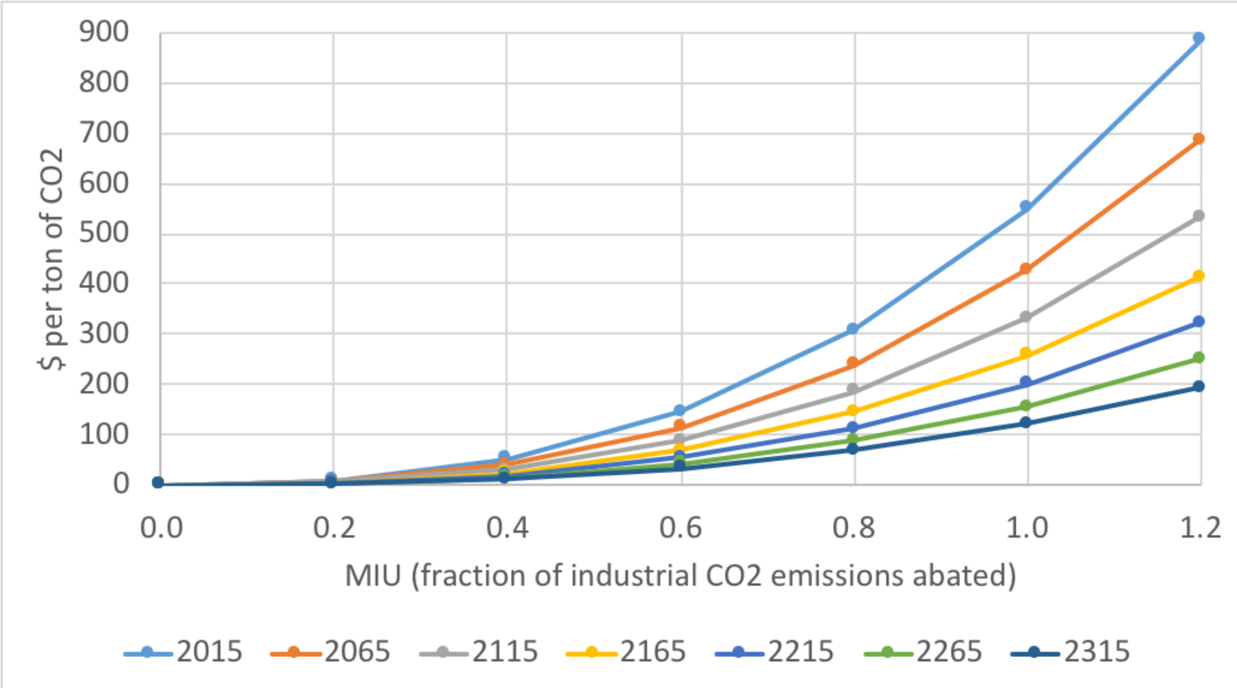

Figure ?: Marginal cost of abatement as a function of fraction of emissions abated, over time

When we look at the absolute amount of emissions abated, however, we get a somewhat different story. Now, we see that the marginal costs for a given level of reduction falls until 2165, but then starts to rise again. The steadily declining cost of the backstop technology steadily drives down costs over time. The level of uncontrolled emissions, though, using the default settings in the model, rises until 2130 the year, after which time it falls. As these rise, the absolute amount of emissions abated for any fractional rate of abatement also increases. This further drives down the marginal cost for each unit of emissions abated, reinforcing the effect of the declining cost of the backstop technology. As these decline, however, the opposite occurs, driving up the marginal cost for each unit of emissions abated and counteracting the effect of the declining cost of the backstop technology. At some point, this second effect begins to dominate, so we see an increase in the marginal costs as a function of the absolute amount of emissions abated.

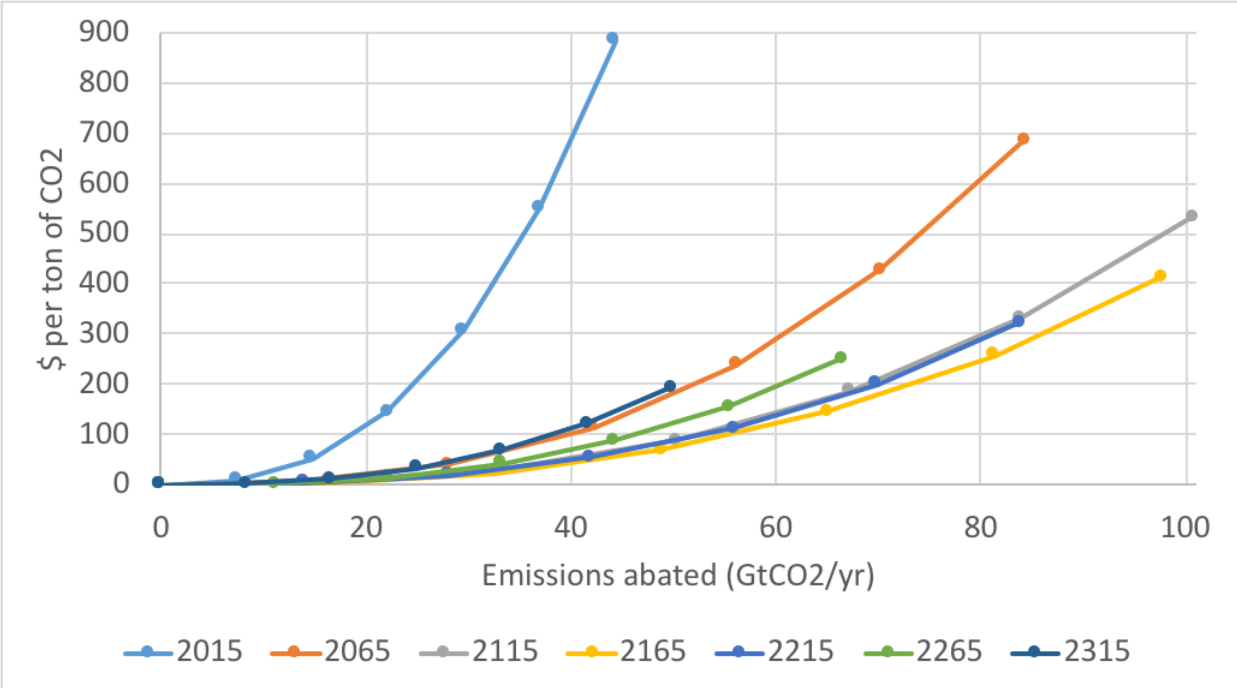

Figure ?: Marginal cost of abatement as a function of level of emissions abated, over time, DICE default values for uncontrolled industrial emissions

This issue becomes more significant when we consider cases with different levels of baseline emissions. Figure ? shows the results when we use the values for uncontrolled industrial CO_2 emissions from SSP1. In this baseline, these emissions remain fairly stable until around 2060, after which time they start to fall fairly steadily. As a result, the marginal costs for each unit of CO_2 abated do not decline as fast as they do in the DICE baseline in the early years and start increasing much sooner; the figure shows this occurring after the year 2065. In fact, in 2065, the marginal cost for any given amount of CO_2 abated is already 5.6 times as high as in the DICE baseline; this ratio increases to over 36 in 2115 and continues to rise thereafter.

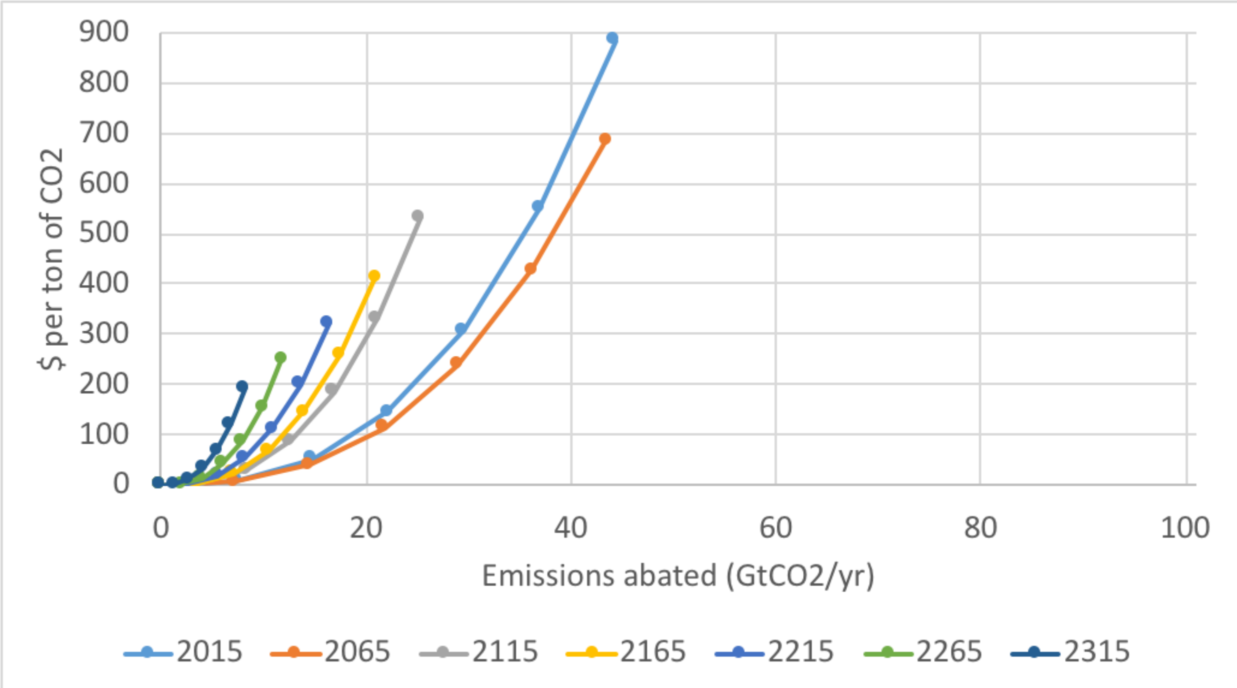

Figure ?: Marginal cost of abatement as a function of level of emissions abated, over time, SSP1 values for uncontrolled industrial emissions

This raises a general question. Does it make more sense for the marginal costs to be determined as a function of the fraction of uncontrolled emissions abated or the absolute level of uncontrolled emissions abated?

Separating Out CDR from Abatement

- **Issue of CDR and mitigation costs**: these are a function of the level of control relative to the amount of uncontrolled emissions; when these emissions are low, e.g. in SSP1 and 4, end up with large MC even for small levels of control, when measured in actually tons of CO_2 abated or removed from atmosphere. **Need to think about how to modify the cost curves, especially for CDR.**
  - Rickels (2018): uses a function, $CDR_{cost} = c1*CDR + c2*CDR^2$, where CDR is in GtC per year, which yields a *linear marginal cost curve*: c1 + 2c2*CDR, where c1 = 0.29328e12 $US/GtC (= 80 $/tCO₂) and c2 varies from 0.01833e12 to 18.33e12 USD/GtC^2
    "CDR is measured in GtC, and c1 in 10^{12} USD/GtC and c2 in 10^{12} USD/GtC^2 . Parameter c1 was set to 0.29328 × 10^{12} USD/GtC, corresponding to 80 USD/tCO2 for the initial amount of (oceanic) CDR (Klepper & Rickels, 2012). For parameter c2 we considered a broad parameter range (to reflect the uncertainty about the scale of CDR), ranging from 0.01833 × 10^{12} USD/GtC^2 to 18.33 × 10^{12} USD/GtC^2 , corresponding to a marginal CDR cost at the first GtC of 90 USD/tCO_2 and 10,080 USD/tCO_2, respectively. The marginal CDR cost curve is linear in the amount of CDR, and increasing with a slope c2."
  - Keith et al (2006): basically a *flat marginal cost curve*; CDR has base cost of $75/tC, which falls over time by 1%/yr; also have multiplier that is a function of

the rate at which CDR (Z measured as fraction of reference emissions) changes, 1 + $(\tau dZ/dt)^2$, where τ is characteristic time of energy systems, given as 50 years; see decline if Z falls; actual amount of CDR given by Z*EmisRef; would probably make more sense to have the adjustment multiplier be a function of the change in actual amount of CDR, and maybe not fall as CDR falls

- Strefler et al (2018, SI): use REMIND, parameters

Table S2: techno-economic parameters of DAC

Investment costs [US$2015/(tCO2/yr)]	100
O&M costs [US$2015/(tCO2/yr)]	2.5
Lifetime [years]	20
Electricity demand [GJ/tCO2]	2
Heat demand [GJ/tCO2]	10

- Fuss et al (2018): "As shown in table 1, in general, cost estimates range from US\$30-\$1000/tCO^2 (p.17)"; "depending on how one chooses to provide energy to the DAC plant will ultimately determine the cost of avoiding CO_2 in the atmosphere. (p.18)"; "Based upon our literature review, it appears that a first-of-a-kind plant may be on the order of US\$600–1000/tCO_2 initially, but that as advances are recognized through the building of more plants, this cost may decrease to US\$100-300/tCO_2 (p.20)"; "Our judgement on potential is 0.5–5 $GtCO_2$ yr^{-1} by 2050 . . . potentials of up to 40 $GtCO_2$ may be possible by the end of the century (p.20)."; no discussion of how marginal costs change with scaling of CDR
- Chen and Tavoni (2013, p.62): use WITCH
- Marcucci et al (2017): use MERGE "We assume that the DAC technology is available from 2060 (following Chen and Tavoni 2013) with an initial cost of 600 US\$ 2010 /tCO_2 captured and a floor cost of 200 US\$ 2010 /tCO_2, covering the ranges in the literature (see Table SOM3). Regarding the thermal energy use, the theoretical minimum is estimated to be around 1 GJ/ tCO_2 (APS 2011; Keith et al. 2006); actual energy use estimations are in the range of 3–10 GJ/ tCO_2. We model

Table 1 Costs and energy consumption of a 1MtCO2 / yr. DAC Plant

	'Realistic' case	'Optimistic' case
Physical Capital	\$350/tCO_2 captured	\$260/tCO_2 captured
Operation and Maintenance (Maintenance, Labour and Consumables)	\$120/tCO_2 captured	\$90/tCO_2 captured
Electricity consumption	490 GWH	490 GWH
High-temperature heat consumption	8.1 PJ	8.1 PJ
CO2 transport and storage cost	Based on regional storage capacity, see Table. A1 for supply cost curves	Based on regional storage capacity, see Table. A1 for supply cost curves

APS (2011)

an increase in the efficiency, with an initial thermal energy requirement of 8.1 GJ/tCO_2 (APS 2011; Chen and Tavoni 2013) and a minimum value of 5 GJ/tCO_2 (Zeman 2007). Moreover, following APS (2011), Baciocchi et al. (2006) and Stolaroff (2006), we consider an electricity consumption of 1.8 GJ/tCO_2 captured.

- Mariia's approach: scale the marginal (and thus total) costs based on size of reference emissions in each SSP baseline relative to the reference emissions in the original baseline in DICE. The idea is to find the level of control in a model with a different baseline that has the same marginal costs.

Start with the MC function for abatement

$$
\begin{aligned}
MCABATE_{s,t}^{i} &= pback_t * \left(MIU_{s,t}^{i}\right)^{expcost2-1} \\
&= pback_t * \left(\frac{EIND_{Base,t}^{i} - EIND_{s,t}^{i}}{EIND_{Base,t}^{i}}\right)^{expcost2-1}
\end{aligned}
$$

where:
i is baseline;
s is policies, i.e., controls allowed, e.g. mitigation, CDR, and SG (s=Base means only base level of mitigation, which set to 3% of industrial emissions);
t is time period;
pback is cost of backstop in \$2010/tCO_2, which currently does not vary by baseline or policy;
MIU is the level of abatement expressed as a fraction of base industrial emissions;
EIND is industrial emissions in GtCO_2/year; and
expcost2 is a coefficient currently set to a default value of 2.6

Let i = 0 represent the 'reference' baseline, e.g. one based on the default parameters in DICE determining pop, tfp, sigma, eland, and forcoth, and i = 1 represent a 'new' baseline, e.g. one based on the SSP values for these parameters.

Calculate: $\beta_t^1 = \left(EIND_{Base,t}^1 / EIND_{Base,t}^0\right)$, which yields $EIND_{Base,t}^1 = \beta_t^1 * EIND_{Base,t}^0$

If we set $EIND_{s,t}^1 = \beta_t^1 * EIND_{s,t}^0$, we have:

$$
\begin{aligned}
MCABATE_{s,t}^{1} &= pback_t * \left(MIU_{s,t}^{1}\right)^{expcost2-1} \\
&= pback_t * \left(\frac{EIND_{Base,t}^{1} - EIND_{s,t}^{1}}{EIND_{Base,t}^{1}}\right)^{expcost2-1} \\
&= pback_t * \left(\frac{\beta_t^1 * EIND_{Base,t}^{0} - \beta_t^1 * EIND_{s,t}^{0}}{\beta_t^1 * EIND_{Base,t}^{0}}\right)^{expcost2-1} \\
&= pback_t * \left(\beta_t^1 * MIU_{s,t}^{0}\right)^{expcost2-1}
\end{aligned}
$$

This would imply that, using the new baseline, marginal costs at control level $\beta_t^1 * MIU_{s,t}^0$ will be the same as marginal costs at control level $MIU_{s,t}^0$ in the reference baseline. Since $expcost2 - 1 > 0$, this means that if the new baseline has lower base-level (uncontrolled) emissions than the reference baseline, i.e., $\beta_t^1 < 1$, the new baseline will have higher marginal costs at any level of control. This would seem to be problematic. To correct for this problem, we can to rewrite the marginal cost function as

$$MCABATE_{s,t}^i = pback_t * \left(\beta_t^i * MIU_{s,t}^i\right)^{expcost2-1},$$

which gives yields the total abatement cost function:

$$TCABATE_{s,t}^i = \frac{YGROSS_{s,t}^i * sigma_t^i * pback_t}{1000 * expcost2 * \beta_t^i} * \left(\beta_t^i * MIU_{s,t}^i\right)^{expcost2}$$

where:
β_t^i, i, s, t, MIU and expcost2 are defined as above;
$YGROSS_{s,t}^i$ is gross output in trillion \$2010/yr; and
$sigma_t^i$ is carbon intensity in tons CO_2 per 1000 2010\$ gross output, which is currently not affected by policy

Note that we can rewrite the new MCABATE function as:

$$MCABATE_{s,t}^i = pback_t * \left(\beta_t^i * \left(\frac{EIND_{Base,t}^i - EIND_{s,t}^i}{EIND_{Base,t}^i}\right)\right)^{expcost2-1},$$

$$= pback_t * \left(\left(\frac{EIND_{Base,t}^i}{EIND_{Base,t}^0}\right) * \left(\frac{ABATE_{s,t}^i}{EIND_{Base,t}^i}\right)\right)^{expcost2-1},$$

$$= pback_t * \left(\frac{ABATE_{s,t}^i}{EIND_{Base,t}^0}\right)^{expcost2-1},$$

where: $ABATE_{s,t}^1$ is the amount of abatement in $GtCO_2$/year
$EIND_{Base,t}^0$ and $pback_t$ (for the moment) do not change when we change baselines. Thus, this last equation implies that the marginal cost of abatement, as a function of the level of abatement in $GtCO_2$/year, will be the same across all baselines, but will evolve over time as $EIND_{Base,t}^0$ and $pback_t$ change over time.

We can similarly rewrite the equations for the marginal and total cost of CDR as:

$$MCCDR_{s,t}^i = pback_t * \left(\beta_t^i * \left(1 + MIUCDR_{s,t}^i\right)\right)^{expcost2-1},$$

which gives yields the total abatement cost function:

$$TCCDR_{s,t}^{i} = \frac{YGROSS_{s,t}^{i} * sigma_{t}^{i} * pback_{t}}{1000 * expcost2 * \beta_{t}^{i}} * \beta_{t}^{i\,expcost2} * \left(\left(1 + MIUCDR_{s,t}^{i}\right)^{expcost2} - 1\right)$$

where:
MIUCDR is the level of CDR expressed as a fraction of base industrial emissions;

This approach gives precedence to the baseline based on the default parameters in DICE that can affect the reference level of industrial emissions. Thus, adjustments should be made when one implements changes in these parameters within DICE.

Finally, we need to give thought to whether: 1) we might want to address the DICE baseline itself by adjusting costs as a function of how the absolute amount of uncontrolled industrial emissions change over time and 2) it would make sense to let the rate at which pback falls (given by gback) vary by SSP. For the latter can look at the rates of change of tfp and sigma in the SSPs relative to the DICE baseline.

- Alternative 1: Similar to Mariia's approach, but create more general way to scale marginal and total costs based on absolute size of reference emissions, so not reliant on original baseline in DICE
- Alternative 2: Uncouple Abatement and CDR cost curves, with the latter being a function of absolute amount of carbon removed
- Alternative 3: adjust the rate of decline of the price of the backstop technology based on the relative rates of decarbonization or tfp growth. Can this be combined with the fix to deal with different levels of baseline emissions?

SCC: Social Cost of Carbon

- defined as (dUTILITY/dETOT)/(dUTILITY/dC) = dC/dETOT; in effect the marginal change in per capita consumption for a 1 ton change in total carbon emissions (changes each year)
- dUTILITY/dETOT = ETOTeq.m
- find that ETOTeq.m = EINDeq.m = -CDReq.m
- good to see that SCC = 0 when no damages
- Expect that CPRICE (= MCCONTROL = max(MCABATE, MCCDR)) = SCC, but
  - SCC > CPRICE, when
    - hit mitigation constraint and no CDR (scenarios Base, M, MG, first year in all scenarios where MIU and MIUCDR prescribed)
    - no mitigation and SCC < reserve price for CDR, i.e. the MCCDR when CDR=0 (scenario C, CG); **note that this is due to how we defined MCCONTROL for cases where CDR = 0. Look back at this.**
  - SCC < CPRICE when:
    - not sure; where expect to see SCC = CPRICE, see, usually small, differences; perhaps just numerics
    - have tried removing constraints on first year and on S at end; when remove constraint on FY only SCC stays < CPRICE without much change in size; when remove constraint on S, SCC initially grows larger than CPRICE, then falls to level slightly below CPRICE until last periods; when remove both constraints, SCC falls well below CPRICE, even going negative for a period, but then SCC exceeds CPRICE for a period, before falling below again
- **Could modify model so that CPRICE is the choice variable and MIU and MIUCDR follow from this.** (See Nordhaus and Stzorc, pp.13-14). Expect that this would lead to case where CPRICE always = SCC, but exceeds MCCONTROL in situations where hit constraints on mitigation and is less than MCCONTROL in situations where SCC < reserve price for CDR and have no mitigation
- What is SCC supposed to be?
  - EPA Fact Sheet (https://www.epa.gov/sites/production/files/2016-12/documents/social_cost_of_carbon_fact_sheet.pdf):The SC-CO2 is a measure, in dollars, of the long-term damage done by a ton of carbon dioxide (CO2) emissions in a given year. This dollar figure also represents the value of damages avoided for a small emission reduction (i.e. the benefit of a CO2 reduction).
  - Nordhaus (2014, p. 273-4): "This term designates the economic cost caused by an additional ton of carbon dioxide emissions (or more succinctly carbon) or its equivalent. In a more precise definition, it is the change in the discounted value of the utility of consumption per unit of additional emissions, denominated in terms of current consumption. In the language of mathematical programming, the SCC is the shadow price

of carbon emissions along a reference path of output, emissions, and climate change."

Understanding what abatement, CDR, and SG do:

- Increase abatement or CDR:
  - Benefits: lower MAT -> lower FORC -> lower TATM -> lower DAMFRACCLIM -> lower DAMAGES -> increase YNETALL -> increase C -> increase CPC; also increase I -> increase K_{t+1} -> increase $YGROSS_{t+1}$; also lower MAT, FORC, TATM, DAMAGES in future years
  - Costs: increase TCCONTROL -> lower YNETALL lower C -> lower CPC; also lower I -> lower K_{t+1} -> lower $YGROSS_{t+1}$
- increase G:
  - Benefits: lower FORC -> lower TATM -> lower DAMFRACCLIM -> lower DAMAGES -> increase YNETALL -> increase C -> increase CPC also increase I -> increase K_{t+1} -> increase $YGROSS_{t+1}$; no direct effect on $FORC_{t+1}$ since no change in MAT_{t+1}; effect on $TATM_{t+1}$ ambiguous as $TATM_t$ smaller, $c1*(FORC_{t+1}-(f2xco2/t2xco2)*TATM_t)$ larger, and $-c1*c3*(TATM_t - TOCEAN_t)$ smaller – first effect would make $TATM_{t+1}$ smaller, second and third effects would make $TATM_{t+1}$ larger
  - Costs: increase GSIDE -> increase DAMAGES -> lower YNETALL lower C -> lower CPC; also lower I -> lower K_{t+1} -> lower $YGROSS_{t+1}$
- In all cases, comparing marginal benefits and marginal costs at a given point in time, without considering future impacts can be misleading

Preparing SSP Results for use as Exogenous Data in DICE

Obtaining the Raw SSP Data

The raw data were downloaded from the © SSP Public Database (Version 2.0) - https://tntcat.iiasa.ac.at/SspDb/dsd?Action=htmlpage&page=40; Accessed on 01/22/2019.

The scenarios selected were:

- IMAGE - SSP1-Baseline
- IMAGE - SSP2-Baseline
- IMAGE - SSP3-Baseline
- IMAGE - SSP4-Baseline
- IMAGE - SSP5-Baseline

The variables selected were:

- For total output (YGROSS): GDP|PPP in billion US$2005/yr
- For population (POP): Population in millions
- Emissions|CO2|Fossil Fuels and Industry in $MtCO_2$/yr
- For land-use emissions (ETREE): Emissions|CO2|Land Use in $MtCO_2$/yr *(ELAND)*
- For carbon intensity (SIGMA): Emissions|CO2|Fossil Fuels and Industry per GDP(PPP) in Mt CO2/yr/billion US$2005/yr
- For other forcing (FORCOTH): Total Forcing (Diagnostics|MAGICC6|Forcing) in W/m^2 and CO_2 Forcing (Diagnostics|MAGICC6|Forcing|CO2) in W/m^2. The latter is subtracted from the former to get non-CO_2 forcing.

The data are provided for the years 2005 and every 10 years from 2010 to 2100.

Interpolating the Raw Data to Annual Values

The raw data were interpolated to annual values using a quadratic B-Spline algorithm. This was implemented using the Interpolations.jl package (v0.11.2) in the Julia software (v1.1.0).

Deriving Implicit Values for Total Factor Productivity (TFP) for the period 2015-2100

The SSPs do not provide explicit forecasts for TFP. However, with the SSP data provided and some assumptions from DICE, we can calculate implied values. Our first effort was to assume no climate damages or any climate policy; rewrite the optimization problem in DICE with the savings rate, S, as the only choice variable; treat TFP as an endogenous variable and total gross production as an exogenous parameter; use the DICE assumption for the capital elasticity in the production function; and adjust the initial value of capital so that the initial value for TFP.

In running the optimization, however, the results were problematic. Since the choice of the savings rate determines both capital investment and consumption per capita, which then determines period utility. The capital investment determines the level of capita in the next period. Since the values for population and capital elasticity are exogenous, the values for TFP must take on the value that gives the right result for total gross production. The net result is that decreasing S in one time period will be balanced by an increase in TFP in the following time period. As S falls toward 0, you get more consumption, and therefore a higher utility. This drives the optimal choice for S toward 0 in each time period; it could be even lower, but that would eventually result in negative values for capital, which is not reasonable. Besides being illogical on the surface, this also results in unreasonably high values for TFP.

Thus, rather than optimizing by the choice of S, we chose to set a constant value for S and then directly calculate TFP. The value we chose for S was the optimal savings rate, as given in DICE. This value, 0.258 is very close to what we see in the regular runs of DICE, so we were comfortable with this choice, and the resulting implicit values for TFP in the different SSPs. These are shown below, along with the values for TFP using the default values in DICE v2016R2.

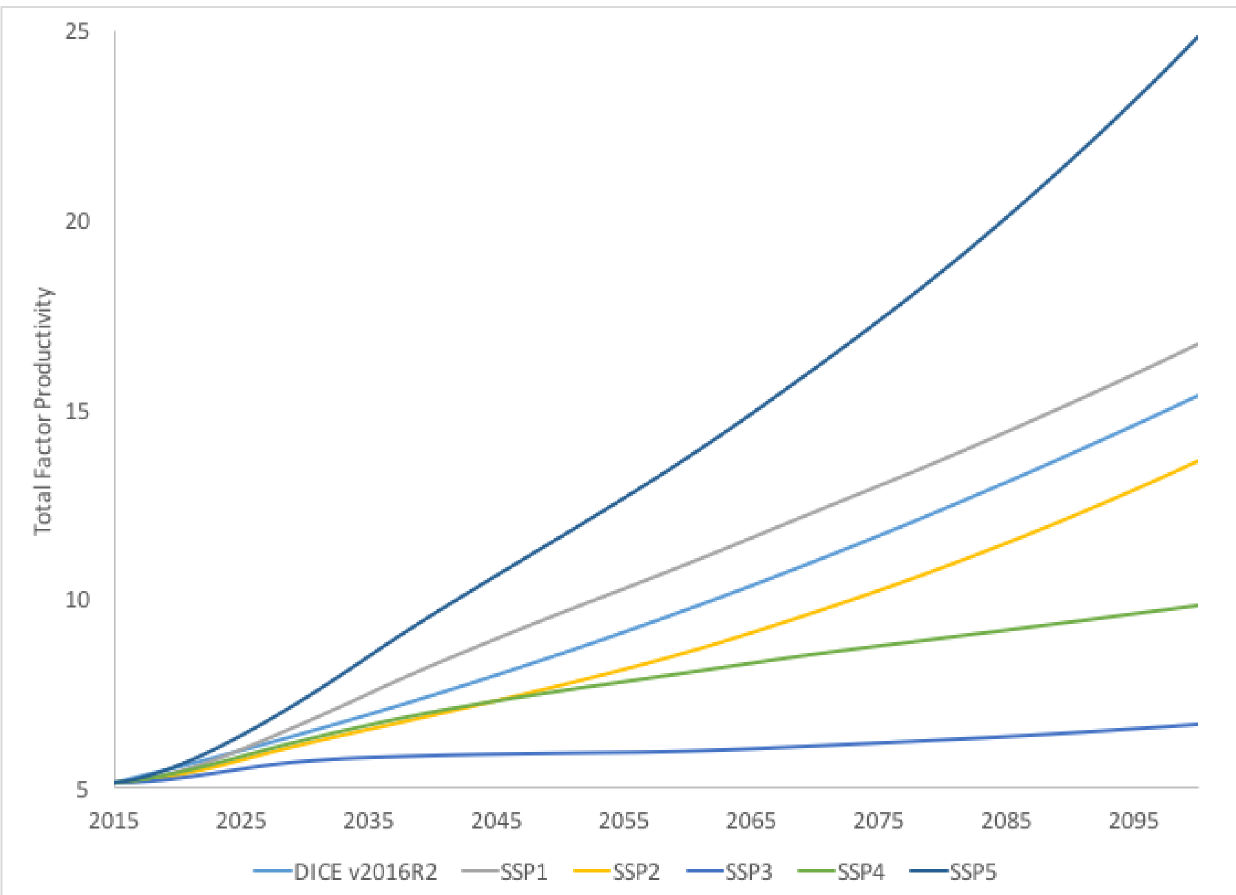

Figure ?: Total Factor Productivity 2015-2100

Extending the Data from 2100 to 2510

As a first step, we normalized the SSP data to the DICE data by multiplying the data in each year by the ratio of the DICE value in 2015 to the SSP value in 2015. This scales the data so that their values match the values in DICE in 2015, but preserves the rates of growth in the SSP data. We then used specific approaches for extending each of the data series from 2100 to 2510.

Population

The SSP data do not follow the same sigmoid pattern as seen in DICE. In fact, in four of the five SSPs, all but SSP3, there is a decline in global population before the end of the 21st century. Since it did not seem realistic for the population to increase or decline in perpetuity, we assumed that they would approach some asymptotic value in each SSP. Furthermore, they approach these values using the same growth parameter used in DICE. Thus, they will follow the path given by:

$$pop_{t+1} = pop_t * \left(\frac{popasym}{pop_t}\right)^{popadj}$$

where popadj = 0.028364 and popasym = 6000, 9000, 13500, 9500, and 7000 for SSP1, SSP2, SSP3, SSP4, and SSP5, respectively.

Total Factor Productivity

The SSP data generally follow the same sigmoid pattern as seen in DICE. Thus, for the post-2100 period, we used the same equation (with a 1-year time step) used in DICE:

$$al_{t+1} = \frac{al_t}{1 - ga0 * e^{(-dela*(t-1))}}$$

Each SSP will have specific values for ga0 and dela, however. To calculate these, we used the same starting value, a0, and assumed that each SSP would have the same asymptotic value as that implied in DICE. This implies that:

$$K = a0 * \left(\frac{1}{1-ga0_{DICE}}\right)^{\frac{1}{dela_{DICE}}} = a0 * \left(\frac{1}{1-ga0_{SSP}}\right)^{\frac{1}{dela_{SSP}}},$$

which can be solved for $ga0_{SSP}$ as a function $dela_{SSP}$:

$$ga0_{SSP} = 1 - \left(\frac{a0}{K}\right)^{dela_{SSP}}$$

For each SSP, we then determined the value of $dela_{SSP}$, and associated value of $ga0_{SSP}$, that provided the best fit to the existing projections to the year 2100. These were, in turn, used to calculate the values of al for the post-2100 period.

Land Use Emissions

The SSPs indicate negative values of land use emissions for all SSPs other than SSP3 by 2100. The values range from -1.9 $GtCO_2$/yr in SSP1 to +0.6 $GtCO_2$/yr in SSP3. For the moment, we simply have these emissions follow a geometric decline toward 0 in all SSPs, where the rate of decline is the same as the default parameter in DICE.

$$etree_{t+1} = etree_t * (1 - deland)$$

where deland = 0.024137 for all SSPs.

This does produce trend breaks in 2100, particularly for SSPs 1 and 5, where the trend shows growing negative emissions even in the period from 2099 to 2100. It may make sense to smooth these out.

Carbon Intensity

Between 2015 and 2100, the SSPs follow a similar declining pattern in the value of carbon intensity, sigma, as seen in DICE, but with a somewhat erratic pattern. Using the implicit value of gsig for the year 2099 and the value of dsig, 0.001, and the equations:

$$gsig_{t+1} = gsig_t * (1 + dsig)$$

and

$$sigma_{t+1} = sigma_t * \ (1 + gsig_t)$$

provides reasonable patterns for all SSPs other than SSP3. SSP3 has very small implicit values for gsig in the last few years, which would imply almost no further decline in carbon intensity. Since this did not seem plausible over the longer-term, we chose to use a slightly higher starting value for gsig in 2099 to produce a more reasonable, but still slow rate of decline in carbon intensity.

Other Forcing

Following DICE, the values for 2100 are used for all values after this date.

Backstop Technology Price

This is not a parameter we could find data for in the SSP database. Thus, for the moment, we have not made any changes related to the price of the backstop technology in the SSPs. In future work, we will consider using different rates of decline, gback, in the different SSPs. These will be linked to the other technological parameters, specifically the rate of growth of TFP and the rate of decline of carbon intensity.

References

Archer, David, Michael Eby, Victor Brovkin, Andy Ridgwell, Long Cao, Uwe Mikolajewicz, Ken Caldeira, et al. 2009. "Atmospheric Lifetime of Fossil Fuel Carbon Dioxide." Annual Review of Earth and Planetary Sciences 37 (1): 117–34. https://doi.org/10.1146/annurev.earth.031208.100206.

Gillingham, Kenneth, William Nordhaus, David Anthoff, Geoffrey Blanford, Valentina Bosetti, Peter Christensen, Haewon McJeon, and John Reilly. 2018. "Modeling Uncertainty in Integrated Assessment of Climate Change: A Multimodel Comparison." *Journal of the Association of Environmental and Resource Economists* 5 (4): 791–826. https://doi.org/10.1086/698910.

Gillingham, Kenneth, William Nordhaus, David Anthoff, Geoffrey Blanford, Valentina Bosetti, Peter Christensen, Haewon McJeon, John Reilly, and Paul Sztorc. 2015. "Modeling Uncertainty in Climate Change: A Multi-Model Comparison." w21637. Cambridge, MA: National Bureau of Economic Research. http://www.nber.org/papers/w21637.pdf.

Nordhaus, William. 2018. "Projections and Uncertainties about Climate Change in an Era of Minimal Climate Policies." *American Economic Journal: Economic Policy* 10 (3): 333–60. https://doi.org/10.1257/pol.20170046.

Nordhaus, William D., and Paul Sztorc. 2013. "DICE 2013R: Introduction and User's Manual, 2nd Edition." New Haven, CT: Yale University. http://www.econ.yale.edu/~nordhaus/homepage/homepage/documents/DICE_Manual_100413r1.pdf.

Nordhaus, William D. 2008. *A Question of Balance: Weighing the Options on Global Warming Policies*. New Haven: Yale University Press.

Nordhaus, William D., and Joseph Boyer. 2000. *Warming the World: Economic Models of Global Warming*. Cambridge, Mass: MIT Press.

Nordhaus, William D. 1994. *Managing the Global Commons: The Economics of Climate Change*. Cambridge, Mass: MIT Press.

Nordhaus, William D. 1992. "The 'DICE' Model: Background and Structure of a Dynamic Integrated Climate-Economy Model of the Economics of Global Warming." 1009. Cowles Foundation Discussion Papers. Cowles Foundation for Research in Economics, Yale University. https://ideas.repec.org/p/cwl/cwldpp/1009.html.

Nordhaus, William D. 1991. "The Cost of Slowing Climate Change: A Survey." *The Energy Journal* 12 (1): 37–65.